

⚠ Educational Prototype — Simulated Data — no real hardware or ocean sensors are connected

• LIVE SIMULATION RUNNING IN YOUR BROWSER

AI Based Ocean Wave Simulation

Smart Ocean Wave Energy System • AI Prediction • Smart Battery • Fault Detection

This project uses AI to study ocean waves and make smart energy decisions. Move the sliders below to change the sea, and watch the buoy, generator, battery and AI respond in real time.

🌊 Wave to Watts

🔋 Battery Aware

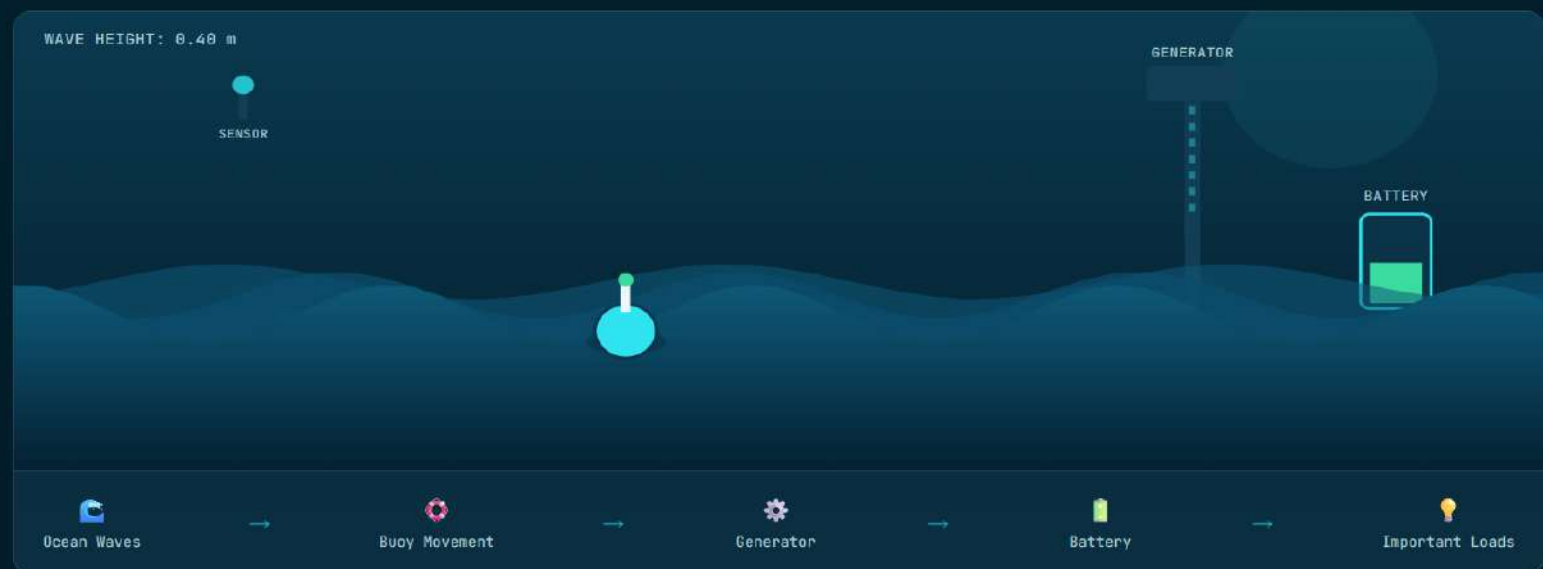
⚙ Priority Loads

🧠 On-device AI Logic

🚨 Coastal Alert & Buzzer

01 Ocean wave energy model

A floating buoy rises and falls with the waves. That movement turns a small generator, which charges the battery that powers important loads.



02 Wave controls

Change the sea conditions and see every reading, chart and AI decision update live.

Wave height

0.40 m

Wave speed

5

Wave frequency

0.20 Hz

▶ Start simulation

⏸ Pause simulation

↺ Reset

Hz = hertz (how many waves pass per second) · m = metres

03 Live sensor data

All values below update every second while the simulation is running.

WAVE HEIGHT

0.40 m

WAVE FREQUENCY

0.20 Hz

WAVE SPEED

5 /10

GENERATOR VOLTAGE

14.9 V

CURRENT

3.5 A

POWER OUTPUT

52 W

TOTAL ENERGY GENERATED

0.0 Wh

BATTERY

50 %

System health: 100%

04 **AI power prediction**

Using the last few readings of wave height, speed and frequency, WaveWise AI estimates power output a little ahead of time.

In 10 seconds

0 W

In 20 seconds

0 W

In 30 seconds

0 W

Confidence

0%

MEDIUM POWER

Start the simulation to see live predictions.

05 **Smart battery control**

The AI recommends one clear action based on the battery level and the power prediction above.

50%

Battery level

Charge Battery

Battery is healthy. Power can be used for normal loads.

Open M

06 Priority load management

Turn loads on or off, or let the AI protect the most important ones automatically when energy is scarce.

Emergency Communication PRIORITY
5 W



Clinic Refrigerator PRIORITY
150 W



Harbour Light
40 W



Water Pump
200 W



Decorative Light
20 W



07 System health and fault detection

Click a demo condition below to see how WaveWise AI reacts to real-world problems.

 Normal System

 Sensor Error

 Low Voltage

 Motor Problem

 Battery Overheating

Health: 100%

Urgency: Normal

Everything is working normally. No action needed.

08 Try a situation

Jump straight to a scenario and watch the whole system react.

 Calm Sea

 Normal Sea

 Rough Sea

 Low Battery

 Generator Fault

09 Coastal alert & high-wave buzzer

WaveWise AI watches wave height and how fast it is rising. If waves get dangerously high, it raises an alert level and sounds a buzzer so people living near the shore get an early warning.



All Clear

SAFE

Wave height is normal. No coastal warning is active. The buzzer will sound automatically if waves rise sharply.

Fishing Harbour

0.8 km from shore

Coastal Village

1.5 km from shore

Beach Clinic

2.2 km from shore

Buzzer sound ☒ On

Test Watch

Test Warning

Test Tsunami Danger

Clear Alert

Test buttons simulate what nearby residents would see and hear. During the live simulation above, the alert updates automatically from wave height and how quickly it is rising.

11 Demo report

 Generate Demo Report

 Download Report

WAVEWISE AI - DEMO REPORT

Generated: 8/24/2026, 2:59:08 PM

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WAVE CONDITION

Height: 0.40 m
Frequency: 0.20 Hz
Speed: 5/10
Predicted condition: MEDIUM POWER

POWER OUTPUT

Voltage: 14.9 V
Current: 3.5 A
Power output: 52 W
Total energy so far:0.0 Wh

BATTERY

Level: 50%
AI recommendation: Charge Battery
Reason: Battery is healthy. Power can be used for normal loads.

ACTIVE LOADS

- Emergency Communication: ON (5 W)
- Clinic Refrigerator: ON (150 W)
- Harbour Light: ON (40 W)
- Water Pump: ON (200 W)

 Open

SYSTEM HEALTH

Health: 100%

Fault status: No fault

Urgency: Normal

Suggested action: Everything is working normally. No action needed.

AI PREDICTION

10s: 0 W 20s: 0 W 30s: 0 W

Confidence: 0%

COASTAL ALERT

Level: SAFE

Status: Wave height is normal. No coastal warning is active. The buzzer will sound automatically if waves rise sharply.

Buzzer: On

=====

Educational Prototype - Simulated Data

12 Ask WaveWise AI

Tap a question to get a quick answer. No microphone needed.

What is the current power?

What is the battery level?

What is the AI recommendation?

Is the system healthy?

Is there a coastal alert right now?

 Open



02 Wave controls

Change the sea conditions and see every reading, chart and AI decision update live.

Wave height 0.80 m Wave speed 2 Wave frequency 0.42 Hz

Start simulation Pause simulation Reset

Open

03 Live sensor data

All values below update every second while the simulation is running.

WAVE HEIGHT
0.80 m

WAVE FREQUENCY
0.42 Hz

WAVE SPEED
2 /10

GENERATOR VOLTAGE
24.4 v

CURRENT
9.2 A

POWER OUTPUT
224 W

TOTAL ENERGY GENERATED
2.4 Wh

BATTERY
50 %

System health: 100%

04 AI power prediction

Using the last few readings of wave height, speed and frequency, WaveWise AI estimates power output a little ahead of time.

In 10 seconds
242 W

In 20 seconds
260 W

In 30 seconds
279 W

Confidence
98%

MEDIUM POWER

Sea conditions are steady. Power output should stay moderate.

Open

09 Coastal alert & high-wave buzzer

WaveWise AI watches wave height and how fast it is rising. If waves get dangerously high, it raises an alert level and sounds a buzzer so people living near the shore get an early warning.



WARNING

Waves are dangerously high. The buzzer is sounding — move away from the beach and low-lying shore areas.

Fishing Harbour

8.8 km from shore

Coastal Village

1.5 km from shore

Beach Clinic

2.2 km from shore

Buzzer sound ☒ On

Test Watch

Test Warning

Test Tsunami Danger

☒ Clear Alert

Test buttons simulate what nearby residents would see and hear. During the live simulation above, the alert updates automatically from wave height and how quickly it is rising.

3:00:19 PM WARNING — Waves are dangerously high. The buzzer is sounding — move away from the beach and low-lying shore areas.



AI Based Ocean Wave Simulation

Manager Monitoring Console

Manager sign in

Access the virtual ocean-wave energy monitoring system.

Email

Password

Sign in to dashboard

College demo account

manager@wavewise.ai / wavewise123

- Overview
- Historical data
- Notifications
- Simulation controls
- Sign out

Educational prototype
Data is simulated locally, not from
live ocean sensors.

Plant overview

8/24/2026, 3:02:14 PM

SIMULATION RUNNING 5 Manager

WAVE ENERGY - MANAGER VIEW

Turn every wave into a smarter decision.

Monitor energy generation, protect critical services, and act on early warnings from one beautiful control room.

Virtual plant

+ AI assisted

Critical-load protection

Simulation mode: values are generated in this browser for demonstration. Connect an IoT controller/API before using this with a real wave-energy installation.

Wave height

0.67 m

Metres (m)

Wave speed

6.1 m/s

Metres / second

Wave frequency

0.41 Hz

Cycles / second

Generated power

293 W

Live output

Battery level

75%

Battery status

Power generation trend

Recent simulated generator output (watts)



AI power prediction

Estimated power based on recent wave readings

Next 10 seconds	294 W
Next 20 seconds	303 W
Next 30 seconds	288 W

AI Based Ocean
Wave Simulation

Manager Console

Overview

Historical data

Notifications

Simulation controls

Sign out

Educational prototype

Data is simulated locally, not from live ocean sensors.

Historical data

8/24/2026, 3:02:41 PM

Stored monitoring history

Snapshots are saved in this browser (local storage). They stay after refreshing this page on the same device/browser.

Download CSV

Clear stored data

186 stored readings

Time	Wave height	Speed	Frequency	Power	Battery	AI prediction	Health
8/24/2026, 3:02:38 PM	0.72 m	6.4 m/s	0.38 Hz	317 W	82%	318 W	Sensor error
8/24/2026, 3:02:33 PM	0.71 m	6.7 m/s	0.39 Hz	326 W	81%	325 W	Sensor error
8/24/2026, 3:02:28 PM	0.67 m	6.6 m/s	0.36 Hz	302 W	79%	302 W	Sensor error
8/24/2026, 3:02:23 PM	0.63 m	6.5 m/s	0.38 Hz	312 W	78%	311 W	Sensor error
8/24/2026, 3:02:18 PM	0.69 m	6.5 m/s	0.38 Hz	316 W	77%	316 W	Sensor error
8/24/2026, 3:02:13 PM	0.67 m	6.1 m/s	0.41 Hz	293 W	75%	294 W	Sensor error
8/24/2026, 3:02:08 PM	0.63 m	6.1 m/s	0.41 Hz	295 W	74%	295 W	Sensor error
8/24/2026, 2:52:09 PM	0.54 m	5.5 m/s	0.37 Hz	258 W	28%	257 W	Sensor error
8/24/2026, 2:52:07 PM	0.46 m	5.5 m/s	0.35 Hz	233 W	29%	232 W	Sensor error
8/24/2026, 2:47:13 PM	0.47 m	5.9 m/s	0.36 Hz	245 W	30%	244 W	Sensor error
8/24/2026, 2:46:13 PM	0.55 m	6.2 m/s	0.34 Hz	268 W	31%	267 W	Sensor error
8/24/2026, 2:45:35 PM	0.48 m	5.8 m/s	0.32 Hz	267 W	32%	265 W	Sensor error

- Overview
- Historical data
- Notifications
- Simulation controls
- Sign out

Educational prototype
Data is simulated locally, not from
live ocean sensors.

Manager notifications

8/24/2026, 3:03:13 PM

SIMULATION RUNNING

0
Manager

Manager notifications

Generated when the simulated system detects a fault, low battery, or major recommendation change.

Mark all as read

System update

Battery recommendation: Balanced operation. Generation and battery level are within a normal operating range.

8/24/2026, 3:02:08 PM • Read

System update

Battery recommendation: Reduce non-essential loads. Battery is falling. Keep priority loads available while monitoring predicted wave power.

8/24/2026, 2:41:19 PM • Read

Action needed

Battery recommendation: Emergency reserve mode. Battery is critically low. Preserve power for clinic refrigeration and emergency communication.

8/24/2026, 2:37:34 PM • Read

System update

Battery recommendation: Save battery. Low battery and weak wave power. Reduce non-essential loads until generation improves.

8/24/2026, 2:33:40 PM • Read

System update

Battery recommendation: Reduce non-essential loads. Battery is falling. Keep priority loads available while monitoring predicted wave power.

8/24/2026, 12:07:13 PM • Read

Warning

Sensor error: Wave sensor values are unreliable. Inspect and calibrate the sensor.

8/24/2026, 11:49:56 AM • Read

